

Genotyping (*Arabidopsis*)

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Woo-Taek Jeon

gDNA Extraction

- 1) Add 30 μL of TPS solution to an autoclaved 1.5-mL microcentrifuge tube
- 2) Add approximately 100 mg of leaf tissue (about the size of a small fingernail)
- 3) Grind the tissue thoroughly using a 1,000- μL pipette tip (or an autoclaved pestle)
- 4) Incubate at 70°C for 25 min
- 5) Add 40 μL of autoclaved distilled water (ADW)
- 6) Vortex thoroughly
- 7) Store at -20°C until PCR analysis
- 8) Vortex the sample, centrifuge at 13,000 rpm for 1 min, and use the supernatant as the DNA template

*TPS solution

Component	Stock concentration	Final concentration	Volume
Tris-HCl, pH 7.5	1 M	200 mM	10.0 mL
NaCl	5 M	250 mM	2.5 mL
EDTA, pH 8.0	0.5 M	25 mM	2.5 mL
SDS	10% (w/v)	0.5%	2.5 mL
ADW			Bring to 50 mL

PCR Mixture

Component	Volume per reaction
2× ThumbTaq PCR Pre-Mix	10 μL
Genomic DNA	1 μL
Primer mix (10 pmol/ μL ; DW:Forward:Reverse = 90:5:5)	2 μL
Autoclaved distilled water (ADW)	7 μL
Total	20 μL

Reagent: 2× ThumbTaq PCR Pre-Mix https://bio-rl.com/product/product.html?category=015024026028&id=TT116-500&mode=view&sort_mode=

PCR Cycle

Initial denaturation: 95°C for 2 min

Amplification (40 cycles): 95°C for 20 sec / 60°C for 40 sec / 72°C for 1 min/kb

Final extension: 72°C for 5 min

Gel Electrophoresis

Gel composition: 100ml of 0.5x TAE buffer + 1.2 g of agarose + 5 µL of EcoDye
→ microwave > 2 min

Loading: PCR product 7 µL & 1kb ladder 3 µL

Run time: 120v 20min / 140v 15min

Primer guideline

How to identify primers to order for Arabidopsis T-DNA insertion-line genotyping

Finding Primers to Order

1. Search the line identifier in the SIGnAL T-DNA Primer Design website. Record the left primer (LP) and right primer (RP) sequences.

SIGnAL T-DNA Primer Design: <http://signal.salk.edu/tdnaprimers.2.html>

2. If accession information, such as a CS stock number, is not shown, search for the parental line through ABRC.

Arabidopsis Biological Resource Center (ABRC): <https://abrc.osu.edu/>

Left-Border (LB) Primers

SALK Lines

Reference database: <http://signal.salk.edu/tdnaprimers.2.html>

Primer	Description	Sequence (5' → 3')
LBb1	LBb1 of pBIN-pROK2 for SALK lines	GCGTGGACCGCTTGCTGCAACT
LBb1.3	Used by the Salk Genotyping Project; often gives improved results	ATTTTGCCGATTTTCGGAAC
LBa1	LBa1 of pBIN-pROK2 for SALK lines	TGGTTCACGTAGTGGGCCATCG
LB_6313R	LB primer for SALK lines	TCAAACAGGATTTTCGCCTGCT

SAIL Lines

Reference database: <http://signal.salk.edu/tdnaprimers.2.html>

Primer	Description	Sequence (5' → 3')
LB1	Positions 418–451 of pCSA110-pDAP101 T-DNAs	GCCTTTTCAGAAATGGATAAAATAGCCTTGCTTCC
LB2	Positions 390–423 of pCSA110-pDAP101 T-DNAs	GCTTCCTATTATATCTTCCCAAATTACCAATACA
LB3	Positions 350–383 of pCSA110-pDAP101 T-DNAs	TAGCATCTGAATTTTCATAACCAATCTCGATACAC

GABI-KAT Lines

GABI-KAT database: <https://www.gabi-kat.de/>

GK Primer Tool: <https://www.gabi-kat.de/db/primerdesign.php>

1. Open the GABI-KAT database and select Simple Search, then open the GK Primer Tool.
2. Search by gene name and select the desired mutant line.
3. Under Primer and WT amplicons, select Show primer details.
4. Locate the T-DNA primer. Use the number at the end of the primer name as the LB primer designation.
e.g.) GABI-o8409 → o2588(GabiLB)